Samin IT-24007

```
impont java.util.concurrent.atomic.AtomicLong;
public class Samin_Thread extends Thread {
static AtomicLong safeCounter = new AtomicLong (0);
static long unsafeCounter = 0;
static boolean threadsafe;
int increments;
long instanceCounter = 0;
Samin_Thread (int increments){
 this.increments = increments;
 }
@ override
public void run (){
for (int i=0; i< increments ; i++){
  if (threadsafe){
safeCounter.incrementAndGet ();
  }
else {
   unsafeCounter++;
```

```
        instanceCounter++;
    }
}

public static void main (String [] args)
    throws InterruptedException {
if (args.length != 3){
 System.out.println (
 "Usage : java Samin_Thread <thread>
   <increments> <true | false> "
 );
 return 0;
}
Samin_Thread [ ] threads = new Samin_Thread
 Safe counter.set (0);
Unsafe counter= 0;
```

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```
for(int i=0; i<n; i++){
threads[i] = new Samin_Thread(increment);
threads[i].start();
}
long nonstaticTotal = 0;
for(int i=0; i<n; i++){
threads[i].join();
nonstaticTotal += threads[i].instanceCounter;
}
long staticCount;
if (threadsafe){
staticCount = safeCounter.get();
}
else
staticCount = unsafecounter;
}
```

```
long expectedCounter = (long) n * increments;
long difference = Math.abs (staticCount -
                            nonstaticTotal);

String percentage;
if (nonstaticTotal == 0) {
  percentage = (staticCount == 0)
        ? "0.00%"
        : "Undefined";
}
else {
  double percent =
  difference * 100.0 / nonstaticTotal;
percentage = String.format ("%.2f%%", percent);
}

System.out.println (" Student: Samin ");
System.out.println (" Threads: +n);
```

```
System.out.println("Increments per thread: "
                    + increments);
System.out.println("Mode: "
                    (+ threadSafe ?" Thread-Safe":
                    "Unsafe"));
System.out.println("Expected count: "
                    + expectedCount);
System.out.println("static count: "+ staticCount);
System.out.println("Non-static total: " +
                    nonstaticTotal);
System.out.println("Absolute difference: "
                    + difference);
System.out.println("Percentage difference
                    + percentage);
    }
}
```